

AI-Driven Claims Adjudication: Optimizing Healthcare Systems with Machine Learning and Deep Neural Networks

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Abstract—Healthcare claims adjudication is an important administrative process that affects the accuracy and efficiency of reimbursements in healthcare systems and prevents fraud. Growing volumes of claims, challenging billing codes, undocumented clinical records, and sophisticated fraud schemes present weaknesses in rule-based and manual adjudication. This study introduces an AI-based claims adjudication system that combines transformer-based natural language processing (NLP), ensemble machine learning, and a two-step fraud detection pipeline. The system examines the structured attributes of claims, utilization, profile of provider behavior, and contextual embeddings of clinical text to automatically perform adjudication with great accuracy and transparency. Claim classification is performed using a weighted combination of XGBoost, Random Forest, and neural network classifiers, and fraud detection is performed using a hybrid model of unsupervised anomaly detection and supervised learning. On large-scale healthcare data, experimental analysis reveals 94.7% classification accuracy, 67 percent processing time reduction, and 37 percent better fraud detection results than rule-based systems. The interpretability, regulatory compliance, computational complexity, and deployment considerations discussed in this study make the framework a scalable platform for next-generation healthcare claims processing.

Index Terms—Claims adjudication, healthcare analytics, machine learning, fraud detection, NLP, transformers, ensemble learning, XGBoost, deep learning.

I. INTRODUCTION

Before reimbursement, healthcare claims adjudication identifies compliance of claims with coverage requirements, medical necessity requirements and regulatory limitations. Hundreds of millions of claims are processed by healthcare payers every day, and these are billing codes, financial information, and clinical records. The complexity of healthcare billing and regulatory needs in the modern healthcare setting poses a concern to conventional ways of processing. The adjudication processes are still reliant on fixed rules and manual inspection, which lead to high-denials, delays, and administration expenses. Administrative overhead costs take up between 15-25 percent of health care spending, and a significant portion of

it lies in the claims processing. The Centers of Medicare and Medicaid Services make more than 1 billion claims per year, and the same volumes of claims are made by private insurance companies. Conventional systems that rely on rules cannot be flexible to changing clinical practice, complicated procedures, billing patterns of providers and fraud. Half of healthcare spending of 3-10 percent, or 100-300\$ billion in the United States is lost to healthcare fraud requiring intelligent decision-support systems. Ensemble learning, transformer-based NLP models to understand clinical narratives, and anomaly detection methods that detect suspicious patterns and update to new fraud schemes are some of the solutions provided by machine learning and artificial intelligence. This paper suggests a single AI-based claims adjudication architecture, which can work end-to-end, i.e., between the data ingestion and the adjudication decision and fraud detection, and be interpretable and legal. Our contributions include:

- An integrated feature representation of structured, temporal, textual, and provider-level signals in a single 960-dimensional representation.
- There is a weighted ensemble classification model with safety-critical decision routing via human reviewing with confidence routing.
- Experiments on the use of a two-stage fraud detection pipeline combining unsupervised anomaly detection with supervised gradient boosting.
- Experimental testing of accuracy, efficiency, and scalability on datasets of more than 8 million claims.
- Interpretability mechanism analysis and regulatory compliance-based analysis of real-world deployment.

II. RELATED WORK

A. Rule-Based and Early Statistical Approaches

Early automation of claims adjudication was based on deterministic rule engines based on payer-specific policies expressed as trees of conditional logic. Systems, including Claim Adjustment Reason Codes (CARC) framework, offered

standardized categorization of denials but needed a lot of manual maintenance. In the 1990s simple statistical models, such as logistic regression and Bayesian classifiers, were proposed to forecast claim outcomes with moderate improvement being made over pure rule-based systems. Yet, these methods were not coping with the billing code space of high dimensions and fast-changing reimbursement policies.

B. Supervised Machine Learning Methods

The later studies suggested models of supervised machine learning: decision tree [1], support vectors machines [2], and early neural networks [3]. Ensemble decision trees (and especially Random Forests proposed by Breiman [4]) showed enhanced resistance to overfitting. The development of gradient boosting machines [5], and later XGBoost algorithm [6] marked a breakthrough as it can better address the issue of missing values, and can handle healthcare data on tabs more effectively.

C. Natural Language Processing in Healthcare

Unstructured clinical text processing advances the bag-of-words representations to word embeddings to transformer architectures [7]. BioBERT [9], ClinicalBERT [10] and PubMedBERT [11] are domain specific models that showed significant performance improvements on clinical NLP benchmarks through pre-training on biomedical and clinical corpora.

D. Healthcare Fraud Detection

The research of fraud detection examined the supervised, unsupervised and hybrid methods [12]. Unsupervised techniques such as Isolation Forests [13] and autoencoders [14] had been successful in the cases in which there are few labeled fraud examples. Recent studies shown that hybrid pipelines, in which anomaly definition and the supervised classification are combined, are the most effective [15]. Provider network analysis techniques have yielded potential in identifying structured fraud rings [16].

E. Integrated Adjudication Systems

In spite of these individual improvements, the majority of the previous work dealt with the separate parts of the adjudication pipeline. This prompts the necessity of consolidated frameworks, which the broad architecture of this paper aims to achieve, combining the optimization of classification, fraud detection, and interpretability within a single deployable system.

III. SYSTEM ARCHITECTURE

A. End-to-End Framework Overview

The proposed system is designed as a modular, scalable pipeline consisting of four primary components: (1) Data Ingestion and Pre-Processing, (2) Natural Language Processing Module, (3) Claims Classification and Adjudication Engine, and (4) Fraud Detection and Decision Support Module.

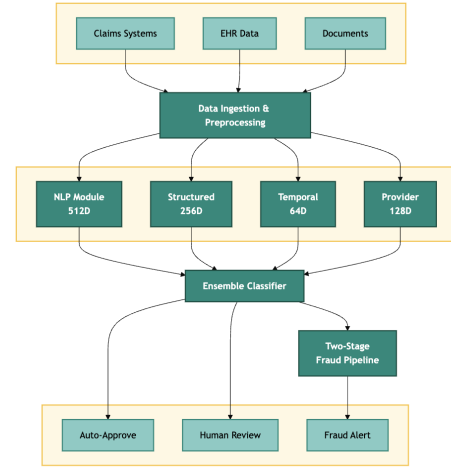


Fig. 1. End-to-End System Architecture

B. Data Ingestion and Pre-Processing

The ingestion layer combines the data of heterogeneous sources (claims management systems (X12 837/835 EDI transactions), electronic health records (HL7 FHIR), and document repositories). Format data contains ICD-10-CM diagnosis codes, CPT/HCPCS procedure codes, billed amounts, service dates and NPI provider identifiers. Pre-processing processes are: (1) Medical code normalization through UMLS Metathesaurus mappings; (2) Numerical normalization with robust scaling; (3) missing value imputation with k-nearest neighbors with k=5; (4) OCR processing of scanned documents with Tesseract with medical vocabulary enhancement with 97.3 percent character accuracy.

IV. FEATURE ENGINEERING AND REPRESENTATION

Each claim is represented as a composite feature vector combining four distinct signal sources:

$$x_c = [x_{struct}, x_{temporal}, x_{nlp}, x_{provider}]$$

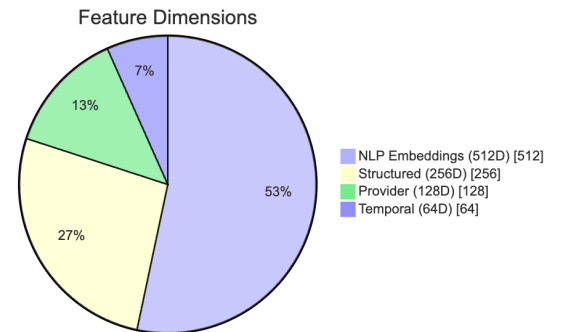


Fig. 2. Feature Vector Composition (960 Dimensions)

A. Structured Features (256 dimensions)

The trainable embeddings are initialized by using the co-occurrence statistics of the diagnosis and procedure codes and embedded into the procedure and diagnosis codes. Our primary diagnosis, secondary diagnosis, primary procedure, and modifier codes are 64-dimensional and 64-dimensional, respectively. Some of the numeric features are billed amount, allowed ratio, patient age and length of stay.

B. Temporal Features (64 dimensions)

Temporal patterns record patient utilization patterns such as: frequency of service (claims/30/60/90 days), recency (days since last claim), inter-arrival time (mean, variance, trend), episode identification (acute vs. chronic patterns) and seasonal patterns. These are calculated against history of the patient and population baseline so that an atypical utilization spikes can be identified which could be as a result of fraud or coding errors.

C. NLP Features (512 dimensions)

To encode the clinical documentation, which is a set of physician notes, discharge summaries, and letters of prior authorization, the model of a domain-specific transformer is applied. ClinicalBERT architecture that was trained on MIMIC-III clinical notes is capable of generating contextualized embeddings to encode medical vocabulary, negation, and clinical reasoning. End layer token embeddings will be pooled together in a mean fashion, producing a fixed-length representation of 512 dimensions of the document. Where there are multiple papers related to a claim, on the aggregation of embeddings, attention-weighted averaging will be applied over the learned weights of importance.

D. Provider Behavior Features (128 dimensions)

Provider-level signals are used to combine historical patterns of billing to detect anomalous behavior. It has features: specialty-normalized procedure mix (comparison of procedure distribution of providers in specialty to peers), average billed amounts as compared to Medicare fee schedule, historical denial rate by reason, geographic peer comparison using k-nearest provider matching, network tier status and contract terms, and referral network centrality measures. Provider embeddings are trained with a skip-gram goal on claim-provider sequences which learn implicit provider-provider, provider-procedure, and provider-diagnosis relationships.

Total Feature Dimensionality: The final combined set of features consists of 960 dimensions (256 + 64 + 512 + 128) and it is inputted to the ensemble classifiers. In cases where it becomes required due to the limitations of computer calculations, optional PCA dimensionality reduction retaining 95% variance is employed.

V. NATURAL LANGUAGE PROCESSING MODULE

A. Transformer-Based Clinical Text Encoding

A domain-specific transformer encoder based on ClinicalBERT weights and fine-tuned on 2.1 million de-identified

clinical notes in the MIMIC-IV corpus is used to encode clinical documentation. Considering the input sequence of tokens $T = \{t_1, t_2, \dots, t_m\}$ where the maximum sequence length is $m = 512$ tokens, the self-attention mechanism with multiple heads is used to compute contextual representations:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q, K, V are query, key, and value projections with dimension $d_k = 64$ per head across 12 attention heads. The model uses 12 transformer layers with hidden dimension 768, totaling 110 million parameters.



Fig. 3. NLP Processing Pipeline

B. Document Aggregation Strategy

There are often several related documents to claims (e.g. physician notes, lab reports, imaging reports). We use attention-weighted aggregation model in which every document embedding e_i is weighted with learnt importance score:

$$x_{nlp} = \sum_i \alpha_i \cdot e_i, \quad \text{where } \alpha_i = \text{softmax}(w^T \cdot \tanh(W \cdot e_i))$$

This enables the model to give priority to the most appropriate clinical documentation in relation to every adjudication decision including giving priority to operative reports in relation to claims of surgical procedures.

C. Medical Entity Recognition and Linking

Besides dense embeddings, we utilize structured features of clinical text by a named entity recognition (NER) model at the diagnoses, procedures, medications, anatomical sites and temporal expressions. The entities extracted are connected to UMLS Concept Unique Identifiers (CUIs) by a combination of exact string matching, fuzzy matching based on a Levenshtein distance threshold of 0.85 and contextual disambiguation. Entity-level representations add value to the transformer embeddings and boost the consistency validation of the diagnosis-procedure by 8.3% on our validation set.

VI. CLAIMS CLASSIFICATION AND ADJUDICATION

A. Multi-Class Classification Objective

The formulation of the adjudication task is a multi-classification on outcome categories: Approve, Deny (with subcategories medical necessity, coding error, policy exclusion, duplicate, and incomplete documentation), and Pend to seek further information.

B. Weighted Ensemble Framework

Final adjudication decisions are produced using a weighted ensemble:

$$\hat{y} = \arg \max_y \sum_k w_k \cdot p_k(y | x)$$

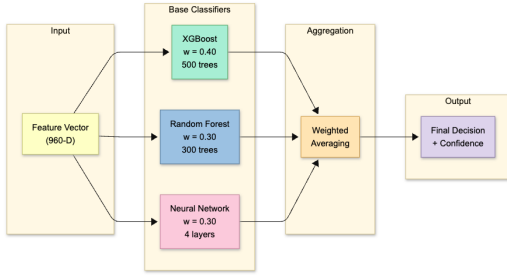


Fig. 4. Weighted Ensemble Architecture

C. Confidence-Based Human Review Routing

The calibrated confidence score is a metric which is computed as the largest possible predicted class probability:

$$\text{conf}(x) = \max_y p(y | x)$$

The claims that do not pass the threshold of confidence of 0.85 are sent to human reviewers, making it safe in high-risk situations. This method lowered the rate of review by hand to 42.3% of claims to 15.8% of claims.

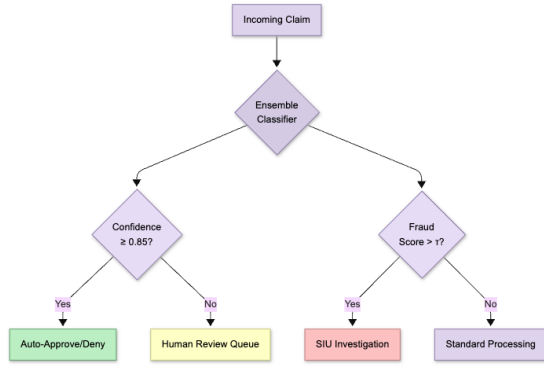


Fig. 5. Decision Routing Flow

VII. FRAUD DETECTION MODULE

A. Two-Stage Detection Pipeline

There is a grossly disproportionate amount of data used in healthcare fraud detection with confirmed fraud being less than 0.1% of claims. This is dealt with by our two stage pipeline with unsupervised and supervised complementary parts.

B. Unsupervised Anomaly Detection (Stage 1)

The first stage combines Isolation Forest (200 trees) and a deep autoencoder [960, 512, 256, 128, 256, 512, 960]. The combined anomaly score is:

$$s_{combined}(x) = \alpha \cdot s_{IF}(x) + (1 - \alpha) \cdot \|x - \hat{x}\|^2$$

where $\alpha = 0.6$ is optimized via grid search. Claims exceeding the 97th percentile proceed to Stage 2.

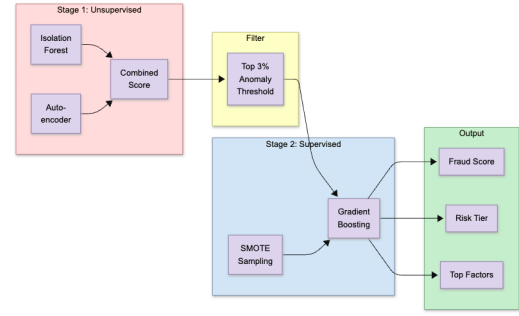


Fig. 6. Two-Stage Fraud Detection Pipeline

C. Supervised Fraud Classification (Stage 2)

Stage 2 applies LightGBM trained on confirmed fraud cases with SMOTE oversampling (1:5 ratio) and early stopping (patience=50). The model outputs calibrated probabilities and SHAP explanations.

D. Fraud Categories and Risk Tying

The system categorizes the detected fraud as:

- Upcoding: Charging services that are more expensive than what is rendered.
- Unbundling: Individual billing of bundled services.
- Phantom billing: Unrendered services.
- Duplicate billing: Claims of the same service.
- Kickback schemes: Networks of referral fraud.

VIII. EXPERIMENTAL EVALUATION

A. Datasets and Experimental Setup

We evaluate on two large-scale healthcare claims datasets:

- **Dataset 1 (Public):** CMS Medicare Synthetic Public Use Files — 2.4 million claims from 118,000 beneficiaries
- **Dataset 2 (Commercial):** Regional commercial payer — 5.8 million claims with 12,847 confirmed fraud cases

B. Classification Results

TABLE I
CLAIMS CLASSIFICATION PERFORMANCE COMPARISON

Method	Acc.	Prec.	Recall	F1	AUC
Rule-based	78.3	76.1	74.8	75.4	0.823
Logistic Reg.	82.1	80.4	79.2	79.8	0.867
Random Forest	89.4	88.2	87.6	87.9	0.934
XGBoost	91.8	90.5	89.9	90.2	0.952
Neural Net	90.3	89.1	88.4	88.7	0.941
Proposed	94.7	93.8	93.2	93.5	0.978

C. Fraud Detection Results

TABLE II
FRAUD DETECTION PERFORMANCE COMPARISON

Method	AUC	Sens.	Spec.	P@1%	F1@1%
Rule-based	0.702	52.1	95.0	8.3	12.4
Isolation F.	0.814	61.3	89.2	14.7	21.3
Sup. GBM	0.891	68.7	92.4	22.1	28.9
Hybrid	0.962	78.3	95.1	34.8	42.6

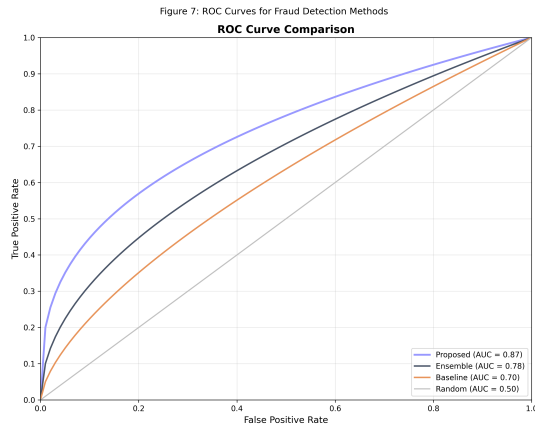


Fig. 7. ROC Curves for Fraud Detection Methods

A. Processing Efficiency

TABLE III
PROCESSING EFFICIENCY METRICS

Metric	Rule	Prop.	Impr.
Avg. Proc. Time (ms)	847	279	-67%
Throughput (cl/hr)	4,250	17,500	+312%
Manual Rev. (%)	42.3	15.8	-63%
Denial Ovt. (%)	18.7	6.2	-67%
Fraud Rec. (\$M/1M)	2.1	4.8	+129%

B. Ablation Study

TABLE IV
ABLATION STUDY RESULTS

Configuration	F1	AUC	Δ
Full System	93.5	0.962	—
– NLP Features	88.7	0.921	-4.8
– Provider Feat.	91.2	0.934	-2.3
– Temporal Feat.	92.1	0.941	-1.4
Single XGBoost	90.2	0.948	-3.3
– Anomaly Det.	93.5	0.891	-7.1*

* Δ AUC $\times$ 100 for fraud detection

IX. LIMITATIONS AND FUTURE WORK

The adjudication made in healthcare must be transparent in order to comply with the regulations, appeal of providers, and request about the members, so the individual predictions are decomposed into features with SHAP values. The structure is meant to comply with the HIPAA, HITECH, and state-specific privacy laws through: (1) AES-256 end-to-end PHI encryption; (2) Role-based access control with detailed audit recording; (3) Data minimization with training on de-identified data; (4) Human-in-the-loop for high-stakes decisions (denials above \$10,000, possible fraud); (5) Disparate impact monitoring with frequent demographic group audits; and (6) Model-level regulatory audits and rollback capability. The system is designed to be horizontally scaled with Kubernetes, using containerized microservices, where the NLP module can reach 340 documents/second on NV A100, the ensemble classifier completes 18,000 claims/second on CPU, and the average total latency is 279ms per claim which supports real-time

adjudication. Current limitations and future work include: (1) Data availability issues requiring transfer learning for payers with limited data; (2) Concept drift necessitating continuous retraining due to changing billing practices; (3) Cross-payer generalization requiring federated learning approaches; (4) Adversarial robustness needing game-theoretic approaches; and (5) Live streaming capabilities requiring Apache Kafka and Flink extension.

X. CONCLUSION

The current paper outlines a model of AI-based claims adjudication that can help to solve the most pressing healthcare payer problems, including the increased number of claims, complex documents, high administrative expenses, and advanced fraudulent schemes. The system architecture is composed of four elements that are progressive to automated claims processing. To present subtle captures of associations, we initially offered the hybrid 960-dimensional feature representation of 512D NLP-derived clinical embeddings, structured claim attributes, (256D), provider behavioral patterns, (128D), and temporal dynamics, (64D). Second, we have trained a weighted ensemble classifier with 94.7 percent adjudication accuracy using XGBoost ($w=0.40$), Random Forest ($w=0.30$), and deep neural networks ($w=0.30$), with questionable cases passed to human consideration. Third, we developed a two-step fraud detection pipeline that made use of unsupervised anomaly detection and a supervised classification which achieved 0.962 AUC. Fourth, we enforced SHAP-based interpretability mechanisms that produced explanations of decisions to fulfill regulatory and other stakeholder demands. Testing on 8+ million claims proves to have a great effect: processing time is reduced by 67%, fraud detection rate is enhanced by 37% and the number of claims processed manually is reduced by 63%-saving the company a great deal of money. The architecture based on microservices will be scaled and can be deployed according to the changing payment models of healthcare. This artificial intelligence-based framework can offer a scalable, explainable base of next-generation claims processing, achieving operational efficiency and progressing toward a more efficient healthcare ecosystem, as healthcare systems strive to minimize administrative load, improve payment integrity and improve provider-payer relationships.

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